

Pawan's Role: Frontend Developer

</> SCROPIDS PLATFORM

Pawan served as the **Frontend Developer** for the ScropIDS project — an AI-powered Intrusion Detection System built to give SOC teams a powerful, real-time window into their security infrastructure. His mandate was clear: transform complex backend data streams into an intuitive, responsive web dashboard that security analysts could actually rely on.

From architecture decisions to pixel-level UI refinement, Pawan owned the entire client-side experience — making ScropIDS not just functional, but genuinely usable.



Made with GAMMA

Frontend Technology Stack

Pawan assembled a modern, production-grade toolchain carefully chosen to deliver a fast, scalable, and maintainable SOC dashboard. Each technology plays a distinct role in the platform's performance and developer experience.

Core Framework

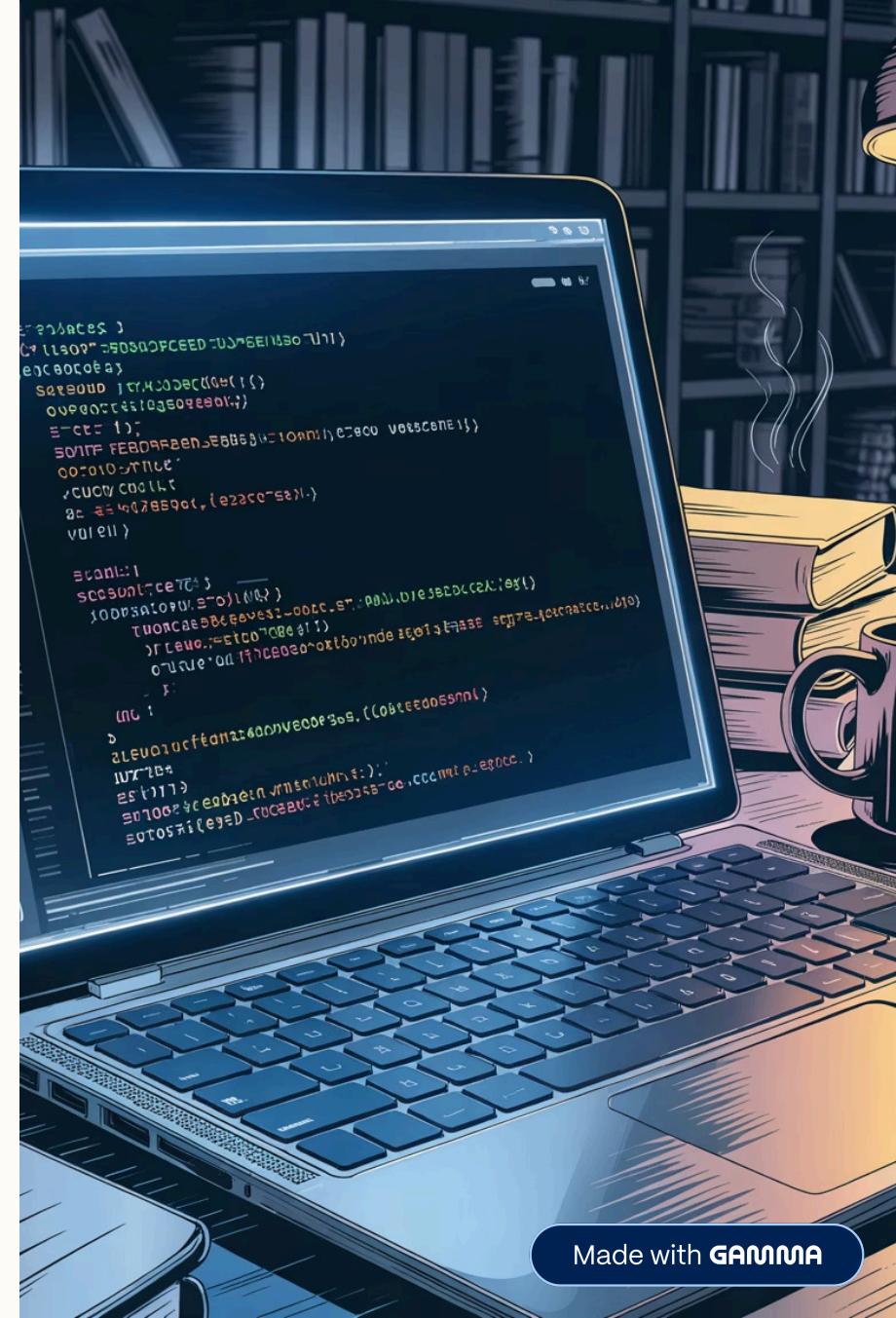
- **React** — component-driven UI
- **TypeScript** — type-safe development
- **Vite** — lightning-fast build tooling

Styling & Navigation

- **Tailwind CSS** — utility-first responsive design
- **React Router** — seamless page navigation
- **Lucide Icons** — crisp, consistent iconography

Data & Visualisation

- **Axios** — reliable HTTP API communication
- **Recharts** — interactive security data charts



Dashboard Development

SOC INTERFACE ENGINEERING

Pawan designed and built every major screen of the ScropIDS dashboard, ensuring SOC analysts had clear, actionable access to all platform data. Each page was deliberately structured to surface the most critical security information with minimal cognitive load.



Overview & Organisations

High-level platform summary showing registered organisations, system health, and active agent counts at a glance.



Agents & Events

Live agent registry with connection status, event logs, and drill-down views for investigating individual endpoint activity.



Alerts & Scheduler

Prioritised alert feed with severity filtering, plus a scheduler settings panel for configuring automated IDS scan tasks.

 All frontend screens are fully connected to backend APIs via **Axios**, ensuring real-time data flow across every dashboard view.

Data Visualisation & Security Insights

Raw security telemetry becomes meaningful only when it is presented visually. Pawan leveraged **Recharts** to build interactive chart components that help SOC analysts instantly grasp the state of their environment – without sifting through raw log tables.

Event Trend Lines

Time-series charts reveal spikes in network events, helping analysts correlate activity with known threat windows.

Alert Severity Counts

Bar and donut charts break down alerts by criticality – critical, high, medium, and low – enabling rapid triage.

Agent Activity Metrics

Live status indicators and activity graphs show which agents are reporting, idle, or disconnected in real time.

Why Visualisation Matters in SOC

Security teams process thousands of events per hour. Well-designed visual components reduce mean-time-to-detect (MTTD) by surfacing anomalies before they escalate. Pawan's chart implementations transformed ScropIDS from a data store into a true monitoring console.

Raw Backend Data

Faster SOC
Decisions

Visual Dashboard
Charts

Final Contribution

PROJECT IMPACT

Pawan's frontend work was the critical bridge between ScropIDS's powerful backend intelligence and the humans who depend on it. By building a structured, performant, and visually coherent dashboard, he ensured that every capability the backend offered was accessible, understandable, and actionable for SOC teams.

Visible

Backend data surfaces as clean UI — organisations, agents, events, and alerts are all visible in structured, easy-to-read views.

Interactive

Users can navigate, filter, configure, and explore — turning a passive data feed into an active security management tool.

User-Friendly

Thoughtful UX design reduces analyst friction, so teams spend time responding to threats rather than navigating the interface.

"The frontend made ScropIDS visible, interactive, and user-friendly."

